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Investigating the role of gamification in public libraries' literacy-centered youth programming

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ABSTRACT
To support youth literacy, U.S. public libraries must constantly evaluate and adjust their offerings to better support how children engage in educational experiences and integrate the latest trends in teaching and learning. This mission makes libraries a unique source of insight into developing perspectives and pedagogy. The authors were curious whether gamification – an increasingly popular design approach – is one of these modern influences. Using online and social media content, the authors developed a list of common youth literacy strategies used by top libraries; this list was then analyzed for elements of gamification. Summer reading clubs, group storytimes, and physical environments stood out as the main initiatives imbued with gamification to better engage and motivate young readers. The goal of the discussion is to help librarians and educators build a foundational knowledge of what gamification is, what it looks like in practice, and what benefits it provides in a learning environment.

KEYWORDS
Gamification; self-determination; youth literacy; public libraries

Introduction

Since opening their doors to young members in the late 1890s, American public libraries have considered children's literacy a primary pillar of their institution. Spurred by an increase in nation-wide literacy rates and a growing focus on childhood reading in elementary schools, public libraries and their affiliated organizations made it their duty to provide dedicated in-house and community resources for young visitors (Hillary & Abbott, 2015). Though many of the initiatives developed during the early twentieth century continue to inform library offerings today, such programs have transformed since their initial inception. One only needs to look at the physical layout of the children's section to see evidence of this development – closed off corners in the basement that once housed the children's stacks (Figure 1) have been replaced with state-of-the-art, multi-level exploration zones, some complete with Willy Wonka type installations, theatrical performances, and even dinosaurs (Figure 2).

How contemporary perspectives in childhood development and learning have fueled these improvements is a worthwhile question, one that is gradually garnering more

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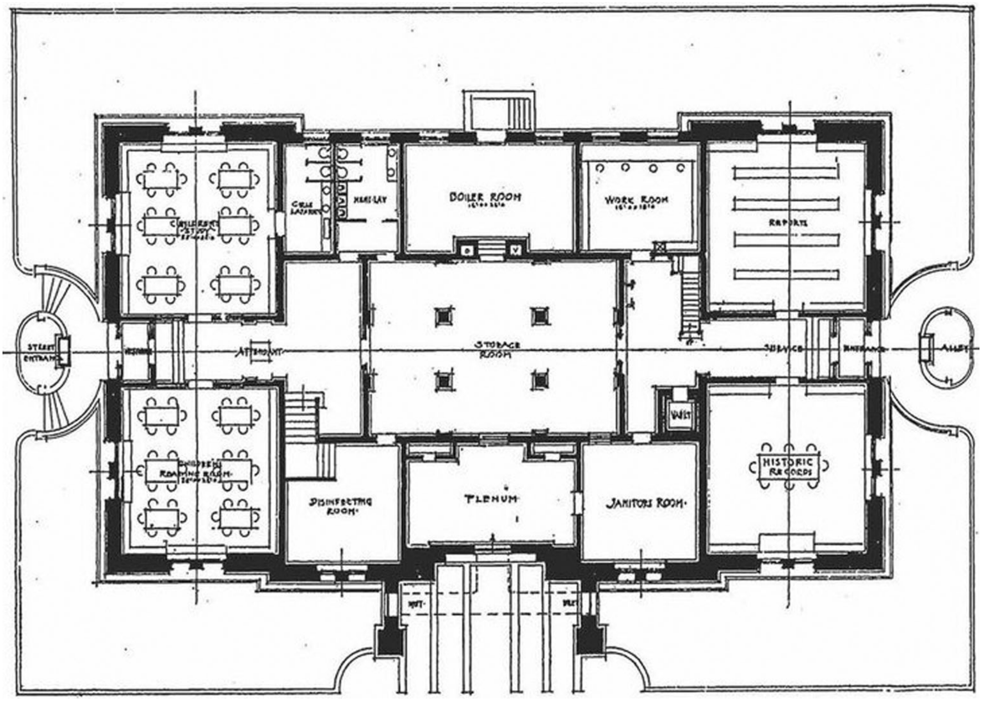

Figure 1. Illustration of Davenport Public Library's basement plans, including a small children's study area and reading room on the same level as the boiler and janitor rooms.
Source: *The Architectural Review*, Volume 4 (1902), p. 25.

attention among researchers, who have, for example, highlighted the impact of early literacy (Teale, 1995), school readiness (Albright et al., 2009; Clark, 2017), and parental involvement (Albright et al., 2009) on the structure of library programming and outreach initiatives. Still, despite a slowly growing body of work on the matter, scholars continue to call for more analysis into the ways our understanding of childhood literacy development has and continues to impact public library strategy (Campana et al., 2016; Jenkins, 2000). To help meet this need, the goal of this paper was to capture which common tools and programs are actively used in libraries today and analyze the public facing language used to portray them. Specifically, the authors sought to determine whether gamification has grown into an area of interest within these institutions. Though often used in the world of business and marketing as a method for increasing user engagement and enjoyment, gamification has become equally prominent in the world of academia as researchers attempt to better define its underlying systems, design, and outcomes across a range of contexts (Nacke & Deterding, 2017). However, because gamification is still a developing research area, there is little existing literature discussing its uses in libraries and related environments.

To help address this gap, this article focuses on the hugely popular and ever-expanding trend of gamification and how it relates to youth-centered library programming. This article first provides a brief literature review to define concepts related to literacy and gamification. It explains what is meant by the terms early and childhood literacy, unpacks the common definition of gamification, and highlights how intrinsic motivation

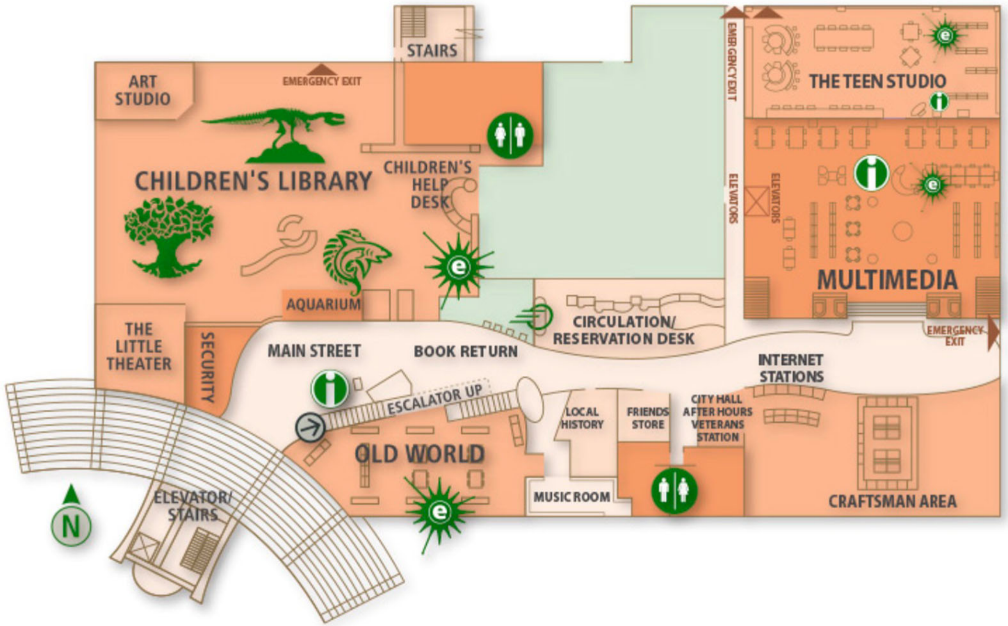

Figure 2. Illustration of the first floor of the Cerritos Millennium Library. Much of the floor is dedicated to specialized children's and teen's spaces, including multimedia, technology, and creative areas.

Source: Cerritos Library online map.

according to Self Determination Theory plays a role in both reading and gameplay. The methodology section outlines the survey approach and findings, after which the article discusses how libraries are incorporating gamification into their summer reading clubs, storytimes, and spaces according to public facing communication and speculates how these approaches may foster youth literacy according to what we understand of gamification's impact on user engagement and behavior. The goal of this analysis is to provide an up-to-date snapshot of library programming that other educators and librarians can use for further research or to inform strategy within their own classrooms and libraries.

Early literacy and motivation

The term 'childhood literacy' may conjured up images of formal instruction on how to navigate words – how to recognize letters, decipher sounds, and scribble lines. Though this style of reading readiness for students above the age of six was certainly the focus for the better part of the early twentieth century, the conversation has since expanded (Teale, 1995). Educators are increasingly adding *early* literacy skills to the conversation, which are the rudimentary building blocks that precede the ability to read but place babies and toddlers in a stronger position to learn in the years to come. Early literacy emphasizes reading and writing behaviors that unfold naturally as pre-kindergarten children engage with books, songs, and other language-rich activities (Urban Libraries Council, 2007). Exactly what these skills are has shifted through the years. While early definitions focused on hard skills, more modern initiatives emphasize an array of literacy-related practices such as singing, writing, and playing in conjunction with well-rounded reading strategies such as oral language and print awareness (Campana et al., 2016).

The purpose of early reading instruction likewise has grown. While reading education was historically intended for, well, reading, many educators now view books and language as tools that can help young children develop the socio-emotional behaviors they need to enter a classroom ready to learn (Teale, 1995). 'School readiness' is the idea that children need to come equipped with healthy and unified mental, physical, and emotional strategies upon entering kindergarten to be successful learners and classmates – and that literacy-based activities can encourage the development of these behaviors (Diamant-Cohen, 2007). Notably, in a 1993 survey conducted by the National Center for Education Statistics (NCES), kindergarten teachers deemed rudimentary skills like counting to twenty as less indicative of school readiness than social and behavioral competency. Over half of NCES survey respondents listed pro-social skills such as the ability to verbally communicate thoughts, follow directions, and take turns as factors that suggest school readiness, while less than 25% rated the ability to identify colors, letters, and numbers as critical to classroom success (as cited by Diamant-Cohen, 2007).

Immersing children in literacy-related activities to build their awareness of reading, writing, and learning strategies and behaviors has an immensely positive impact on their future development (Urban Libraries Council, 2007). However, while there are well-documented benefits to children spending time with books and related content in and out of the classroom, there is, at the same time, a notable lack in their willingness or desire to do so. In fact, the amount of time elementary students spend reading for pleasure is reportedly on the decline (Naeghel et al., 2016). Proponents of Self-Determination Theory (SDT) argue that the cause for this growing ambivalence lies in the motivational relationship between children and text. According to SDT, individuals feel internally motivated to continue engaging in an act when it meets their psychological needs for autonomy, competence, and social relatedness. Respectively, these terms refer to the sense that your actions are your own and are not controlled by an outside influence, that you know what you're doing and your actions have consequence, and that you have a social bond of connection and belonging between yourself and others (Naeghel et al., 2016).

When these needs are met, individuals – or in this case, readers – want to continue the activity because they find it personally interesting, a phenomenon known as intrinsic motivation. This increase in internal motivation improves the child's positive relationship with book-related content, which leads to greater frequency of leisure-time reading, more engagement, great learning and comprehension, and greater enjoyment (Naeghel et al., 2016; Yoon, 2002). Extrinsic motivation, on the other hand, occurs when individuals perform an action in response to external pressure or reward. This type of motivation, created by grades for example, can reduce students' sense of autonomy and cause them to feel so threatened 'that they never experience the pleasure of reading' (Yoon, 2002, p. 188). Thus, it's critical to introduce reading strategies that foster intrinsic participation over its counterpart. Studies have shown that activities such as independent reading with teacher support and structure (Naeghel et al., 2016); peer-to-peer discussions before, during, and after individual reading time (Sanden, 2014); and holding an open dialogue between teachers and students in real-time instead of separate, anxiety-inducing accountability measures such as exams (Yoon, 2002) are able to support autonomy, competence, and social relatedness in ways that result in intrinsic motivation.

Defining gamification

The term gamification has been steadily growing in popularity in recent decades, and today can be found across nearly every sphere of industry and academia (Nacke & Deterding, 2017). According to his own writings, game developer Nick Pelling officially coined the word in 2002 when he used it to describe his process of applying user interface designs commonly found in games to other electronic devices, such as vending machines and phones, to simplify the user experience of using them (Pelling, 2011). Of course, he wasn't the only developer around this time beginning to question how to leverage elements of games and gameplay for purposes beyond entertainment. One only needs to look at the burgeoning children's educational video game surge of the late 90's and early 2000s – with titles such as *Reader Rabbit* and the *Clue Finders* produced by The Learning Company – to see evidence of this growing interest. Still, it is only in recent years that academic researchers have begun to seriously consider gamification's definition, construction, and uses. According to a document repository developed by the makers of JSTOR, in 2011, thirteen documents were published in JSTOR or Portico journals containing the term gamification; in 2020, this number grew to 5,953 total documents, with 1,207 published in 2020 alone (Constellate, n.d.).

But what does the term gamification mean? Often, the word is boiled down into this most basic definition, which is the act of applying gaming elements to non-game contexts to increase user motivation toward a specific outcome. However, gamification is far more complex than a single sentence description as it can applied in multiple ways across a wide range of contexts for an equally vast number of purposes. To better understand the term, it's helpful to unpack each part of its basic explanation, starting with gaming elements. In other words, what is a game made of? While there is still no single consensus to answer this question, there are a few components that frequently appear in suggested frameworks (Seaborn & Fels, 2015). These are game mechanics (the functional and feedback related mechanisms on which the game is built), game dynamics (the actions players perform), and the emotional and behavioral impact on players (Barata et al., 2017). For example, introducing collectibles as a game mechanic may encourage the action of exploration, which may in turn fuel the player's sense of achievement. To help ensure these components are woven together in a coherent way that leverages the strengths of gameplay, researchers have sought to define a series of best practices. Table 1 summarizes some common suggestions.

By leveraging these elements and best practices, games have a way of motivating players in incredible ways – from spending hours cultivating a virtual farm to combating a difficult boss over and over until one finally wins. The key to this persistence should now sound familiar – Self Determination Theory. Just as researchers in the field of literacy are wondering how to encourage students to *want* to read, a major goal of gamification research is better understanding how games and gameplay foster the same intrinsic motivation in players. For example, in a study conducted by Cruz et al. (2017), the authors sought to determine how visual rewards influenced internal and external motivation and how this difference impacted player behavior. Players who viewed these rewards, called badges, as prizes for achievement experienced boosts to their self-esteem, a desire to continue playing, and a heightened sense of skill. Players who viewed the badges as external assignments meant only to control their behavior, however, thought the tasks were meaningless distractions to the game, designed only to trick players into spending their time and money.

Table 1. Common game design best practices and guidelines (adapted from Rapp, 2017; Simões et al, 2013).

Guideline	Description
Repeated experimentation	Players should be allowed to experiment and try different approaches to reach a goal; failure should be used as a positive learning opportunity, not a negative punishment
Rapid feedback	Immediate feedback should be provided to help players improve their strategy and increase their chances of future success
Flow	An ideal game should reach a 'flow' state, which balances a player's skill level and the game's level of difficulty, ensuring that play is neither too easy (which is seen as boring and too little of a challenge), nor too difficult (which is seen as anxiety-inducing and disheartening)
Sub-tasks	Complex goals should be broken into shorter and more simple sub-tasks to help players deal with complexity in a divide and conquer approach
Choice and freedom	Players should be able to choose a different sequence of sub-tasks, following his/her own preferred route to complete the task
Recognition and reward	Player's efforts and accomplishments should be acknowledged
Identity	Players should be encouraged to try new identities and roles to experience and express multiple sides of themselves, as well as develop a sense of social connections

Gamification in public libraries

The last component of gamification focuses on the impact of games, asking what behavioral, emotional, or even cognitive influence the experience may have on its players. These outcome-related goals differ depending on the target industry, but they often relate to increasing user engagement, constructing a social network, encouraging user exploration and experimentation, and so on. For the purposes of this paper, the authors sought to find how existing literature has leveraged gamification in public libraries and for what aims. Using game-based techniques to help patrons navigate the potentially overwhelming physical layout and the process of searching for and requesting materials, as well as a general need to encourage more people to come to the library in the first place, were often discussed (Brigham, 2015; Felker & Phetteplace, 2014 ; Walsh, 2014). However, there was a limited amount of literature that targeted patrons elementary age or younger, and that focused on reading obtainment as opposed to other types of literacy such as digital or information literacy. The few articles found discussed the popularity of gamified apps for summer reading clubs (Kirsch, 2014) as well as other competitive activities such as scavenger hunts that rely on badge-based rewards as their form of motivation (Kim, 2015). The lack of available literature demonstrates a critical need for more research into ways to leverage gamification for childhood literacy.

Materials and methods

The goal of this research was two-fold: first to survey U.S public libraries across the nation to gather common youth literacy initiatives between them, and then to analyze the public presentation of these strategies to determine if and how tenants of gamification are informing their creation and delivery. In doing so, the authors hope to provide an up-to-date view into how libraries are supporting youth education using contemporary perspectives on how children learn, the skills they should acquire, and the role intrinsic motivation plays in this process.

It is worth noting that the age group included in this survey was based on how libraries themselves defined their target audience. While some libraries used terms like ‘kids’ or ‘children’ to refer to all patrons elementary age and younger, others provided further categories such as babies and toddlers. This survey and resulting discussion take the stance of including all pre-k and elementary ages.

The sections below describe the survey approach and findings.

Identifying libraries

Of course, analyzing every library within the U.S would be both daunting and unhelpful; to narrow down the scope to a manageable level, the authors sought to include only those libraries with the most highly ranked children’s divisions. However, there doesn’t appear to be an official children’s library list, so the authors had to rely on other types of materials to develop a ranking system. The first step in developing this criterion was to determine which libraries as a whole – meaning children’s as well as teen, adult, and other spaces – are considered successful and engaging, as these libraries likely have the resources to develop innovative programming. The survey used the longstanding Library Journal's (LJ) Star Library system for this purpose, a program that awards point-based stars to libraries based on their total circulation, circulation of electronic materials, library visits, program attendance, and public internet computer use (Lance, 2020). Libraries that earned 4–5 stars from the Library Journal's 2019 ranking were placed on a long list.

To gain a degree of insight into children's programs specifically, the authors turned to online blogs, travel posts, and articles. These casual, socially-focused writings provide a more 'on the ground' opinion of what constitutes a successful children's space, and their popularity with guardians, tourists, and other potential patrons means they have some influence on where individuals may feel more eager to visit and engage. Writings from Kidpass (Kelly, 2015), TinyBeans (Warkentin, 2019), Brightly (Taylor, 2019), and Book Riot (Jensen, 2018) were used for this stage. These sources were initially selected due to their prominence in web search results. To validate their public regard, the authors looked toward the libraries themselves and whether these institutions applauded their inclusion in these posts. For example, Arlington Public Library published an announcement celebrating their position in Book Riot's top 15 ranking, while a Brentwood city newspaper wrote a similar congratulatory announcement when Brentwood Public Library was ranked by Kidpass.

Libraries listed in both categories – an LJ 4–5-star recipient and a blog's top favorite – were included in the short list. However, a bigger pool was still necessary. The authors briefly analyzed children's libraries frequently listed in the online articles. If a library that appeared repeatedly in the blogs offered unique or modern programming that stood out from the norm, it was included in the survey even if it was not ranked as a LJ star recipient, though several of these libraries have been acknowledged by the LJ as top library of the year (a separate award designation). The Library of Congress and the Carnegie Library were also included due to their unique position and history within library programming and development. Table 2 summarizes each library included in the survey and its reason for selection.

Results

Emergent strategies

The next step was to investigate each library to uncover common trends and approaches between them pertaining to youth literacy. This stage took place online due to the goals and timing of the survey. One major aim of this research was to analyze how libraries present their initiatives on public-facing outlets to gain a sense of what qualities they deem important to highlight for children, guardians, and other members of the public. Additionally, this survey took place during the height of the COVID-19 pandemic when nearly all physical library locations were closed. Due to these aims and considerations, websites, social media posts, and other online materials were thoroughly analyzed to develop a list of literacy-related in-person and virtual offerings, such as events, materials, challenges and competitions, workshops, and other programming designed specifically for children and their caretakers. Table 3 summarizes the most common strategies uncovered during this survey, along with

Table 2. List of U.S. libraries included in the survey and their reason for selection.

Library	State	Top children's library	LJ star ranking[e]	LJ Library of the Year Award[f]
Cerritos Millennium Library	CA	Yes[a,b,d]	5 star	–
Cincinnati & Hamilton County Public Library	OH	Yes[b]	5 star	–
Seattle Public Library	WA	Yes[b]	5 star	–
San Francisco Public Library	CA	Yes[b]	4 star	2018
Children's Library Discovery Center (Queens Public Library)	NY	Yes[a]	–	2009 (Queens PL)
Clinton-Macomb Public Library	MI	Yes[d]	–	–
Hillary Rodham Clinton Children's Library	AR	Yes[b]	–	
ImaginOn: The Joe & Joan Martin Center (Charlotte Mecklenburg Library)	NC	Yes[a,b,d]	–	1995 (Charlotte PL)
John *P.* Holt Brentwood Library	TN	Yes[a,d]	–	–
Laramie County Library	WY	Yes[a,b]	–	2008
Los Angeles Public Library	CA	Yes[b]	–	–
Thomas Hughes Children's Library (Chicago Public Library)	IL	Yes[c]	–	
Elmhurst Public Library	IL	–	5 star	–
Upper Arlington Public Library	OH	–	5 star	–
Carnegie Library of Pittsburgh	PA	–	–	–
Library of Congress	DC	–	–	–

[a]Kidpass.
[b]Tinybeans.
[c]Brightly.
[d]Book Riot.
[e]List found at https://www.libraryjournal.com/?detailStory=ljx191230StarsByNumbers.
[f]List found at https://www.libraryjournal.com/?page=Past-Winners.

some notes regarding each. This list was then used to determine if certain branches incorporated a new trend, technology, or approach that set their design apart from the norm.

Discussion

While analyzing the list of arising themes, the authors noted a clear effort on the part of libraries and their stewards to transform their literacy-based offerings in ways that would better connect with today's young digital natives and make their experiences within the library more authentic, relevant, and enjoyable. Summer reading clubs, storytime hours, and the design of physical space stood out as the main areas that are currently being revamped and approached in new ways. Interestingly, the way in which libraries are innovating these three initiatives to better motivate and support children's reading and language align closely with similar efforts in the realm of gamification. Using SDT's definition of intrinsic motivation as the guide, this discussion aims to unpack the relationship between these programs and gamification to provide examples of how this framework can be integrated into real-world learning contexts. Special attention is paid to the three main psychological needs of intrinsic motivation which, to reiterate, are a sense of autonomy, competence, and social connection. Future educators and librarians may use this information to construct their own activities with a better understanding of how to support learning and motivation through game-based practices.

Due to the unique timing of this paper when libraries were closed, along with the authors' aim of analyzing publicly available materials, the following discussion leverages websites, social media, and other online material to construct an understanding of in-person programming. These sources were also combined with observations of online

Table 3. Emerging library trends to target childhood literacy.

Strategy	Description
Children's websites	*Question raised*: How are websites used to attract, motivate, and provide resources for young children and their caretakers? *Statistics:* • 69% (11/16) offer a children's section or tab nestled within the main website • 19% (3/16) offer a separate children's website, but it mimics the visual style of the main website and/or may be too complex for kids to navigate on their own • 12% (2/16) offer a separate children's website that is visually and functionally designed to appeal to kids
Children's physical spaces	*Question raised*: How are libraries constructing their children's section to attract young patrons and support their learning styles? *Statistics:* • 25% (4/16) provide what might be considered a 'basic' children's section nestled within a small area of the main library with age-appropriate books, decorations, and computer stations • 63% (10/16) go beyond the 'basic' to deliver larger and more customized children's spaces that provide specialized programs, technology, items, and rooms to target literacy outcomes • 12% (2/10) are entire buildings or stores dedicated to youth library services
Storytelling	*Question raised*: What groups are included in story time offerings and what types of skills or activities do these sessions promote? *Statistics*: • 100% (16/16) clearly offered at least one type of in-person storytime • 81% (13/16) clearly offered at least one type of virtual storytime *Insights*: Storytimes were further broken down to address the needs of unique learners, including by: • Age group, including babies, toddlers, elementary students, and so on • Cognitive or socio-emotional concern, such as providing a therapy dog to reduce anxiety during reading • Arts-based resource or activity, such as those that encourage dancing, creating music, yoga stretches, and so on
Summer reading clubs	*Question raised*: How and why do modern summer reading programs differ from those of decades past? *Statistics:* • 100% (16/16) offer at least some sort of competition, incentive, or program for children to keep track of reading during the summer • 50% (8/16) offer activities and worksheets that go beyond a basic reading log to also encourage kids to explore, get involved in the community, and/or complete STEAM based activities • 50% (8/16) used Beanstack, a third-party app, for participants to keep track of progress • 81% (13/16) offer clearly designated prizes based on completed reading minutes, such as an artist tote or gift card
Book kits	*Question raised*: What materials can children and their caregivers take home/check-out to support at-home reading efforts? *Statistics*: • 44% (7/16) offered some form of reading and language kit to take home, many of which include materials, instructions, and games to reinforce literacy skills or to meet unique reader needs such as braille • 19% (3/16) offered book club kits that include novels for participants and discussion questions

(*Continued*)

Table 3. Continued.

Strategy	Description
Workshops	*Question raised*: What type of scheduled activities are offered by libraries to engage young patrons in reading and language learning? *Insights:* • Literacy-based workshops that don't fit into any of the other specified categories include: ○ Tutor-focused programs, such as FOG Readers and Every Child Ready to Read @ your library ○ Reading buddies
Community outreach	*Question raised*: How are libraries extending their literacy-based efforts to help educate the larger community? *Insights*: • Strategies include: ○ Information campaigns to help teach parents and caregivers how to better support literacy learning ○ Partnerships with national organizations such as Reading Recovery or local schools
Recommended 3rd party digital resources	*Question raised*: What online resources do libraries recommend to support at-home learning? *Insights*: • Long bullet-style lists were the most common way for libraries to organize their recommendations, which were long and difficult to navigate • The most recommended apps were e-reader repositories, though some of these, such as TumbleBooks, include animations, narrations, and other fun educational features. • Homework help sites were also frequently listed, and often included interactions, games, and stories • Some apps, such as ABC Mouse, are entirely game based, though still focused on literacy and language

offerings to gain a better understanding of librarian techniques and participant learning goals.

Summer reading programs and rewards

Libraries have been hosting summer reading groups since the turn of the twentieth century to encourage students to continue reading during their extended school break using reading logs and prizes (Bertin, 2004). And they continue to be used widely today, with 95% of libraries offering this type of programming even by 2000 (Celano & Neuman, 2001). The main goal of these programs is threefold: to support children as they develop their reading and language skills; to prevent the 'summer slump', a term describing the common phenomenon of losing academic progress during the break; and to combat aliteracy, which is the ability to read but the lack of desire to do so, often because reading seems, well, boring. Whether prizes are based on minutes read, days logged, or as a final reward for completing all consecutive months of the challenge, these incentives are meant to not only keep track of a child's participation, but also to encourage kids to make reading and related learning activities a continuous priority. An analysis of summer reading programs offered by Pennsylvania libraries found that children who attended these leagues read significantly better than their peers who did not participate in any library groups (Celano & Neuman, 2001).

All libraries included in this survey offered some sort of summer reading challenge or set of activities; of these, half used the Beanstack app to gamify their summer reading challenges. Beanstack is an online user portal that displays a participant's progress, rewards, and to-do lists, all shown in an aesthetically simple and cartoon-like format. For example, activities in ImaginOn's Summer Reading program are categorized into sections called Create, Explore, Give, Play, and Write. To see which goals are in each of these paths, a child clicks on a graphic tile to pull up an interactive checklist. When he completes a task and checks it off, the app provides a small audio and visual reward, and tells the child how many points he has earned (Figure 3). When he completes the entire checklist, the category badge becomes colored, giving a sense of accomplishment but also signaling which badges remain gray (Figure 4). After accumulating 600 points by logging reading hours and/or completing learning activities, the child then earns a final, large reward, such as a new book.

Due to their ease of use, points-based rewards such as badges are the mostly commonly used form of gamification. However, their impact on user motivation is somewhat debated. On one hand, badges set clear goals. They provide visual and textual cues, information on how to fulfill the challenge conditions, and narrative elements to tie the components together (Hamari, 2017). Badges and rewards are thought to increase the quantity and quality of user performance by setting higher expectations that give players a greater sense of self-efficacy when they complete a given task. When these game mechanics are combined with immediate feedback – such as the chipper ding that rings when you finish a checklist item in Beanstack – users feel a greater sense of accomplishment and satisfaction. However, badges can have a negative impact on

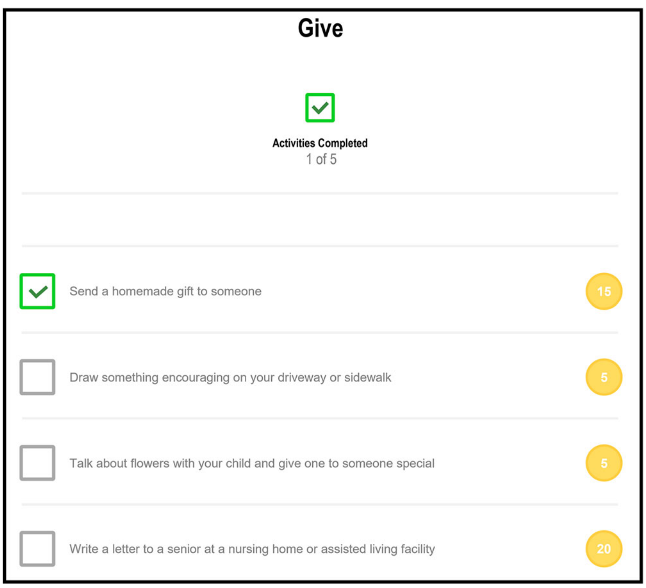

Figure 3. Screenshot of Beanstack interactive checklist, as seen in ImaginOn's Summer Reading Challenge portal.

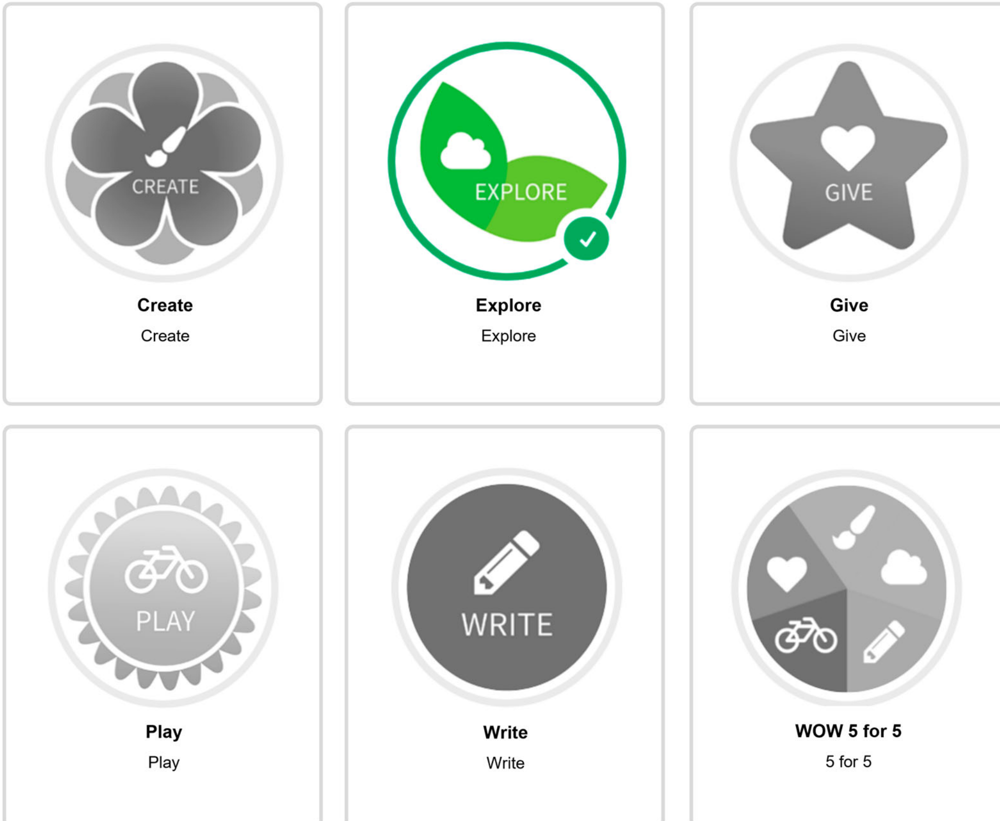

Figure 4. Screenshot of Beanstack complete and incomplete categories, as seen in ImaginOn's Summer Reading Challenge portal.

motivation and engagement if they are perceived as controlling or do not align with the user's needs (Mekler, 2017).

Library summer reading apps such as Beanstack have several qualities that seem likely to prompt more positive reactions from users. Badges in the app are typically given when a child accomplishes an educational activity, engages with the community, or participates in a hands-on learning project. In other words, the rewards are closely tied to personal and social values and often take time and effort to earn. These qualities are thought to make rewards more intrinsically meaningful (Rapp, 2017). Additionally, with its series of checklists, visual iconography, and clearly organized goals, Beanstack gives readers a quick and easy sense of where they came from, where they are, and where they need to go. Not only does the act of recording and reflecting on one's action build a habit but having a sense of an ideal future goal is shown to motivate users to put their best effort into achieving it (Rapp, 2017).

Though many of the app's features support intrinsic motivation, there is one element potentially missing according to Self-Determination Theory: social relatedness. No Beanstack portals analyzed in this survey included features to connect a reader's progress to those of his peers: no leaderboard, no sharing of progress, no chatroom to discuss successes and stumbling blocks. Though this may be due to valid issues such as privacy or liability concerns, it is still important to note as the bond built between participants in a gamified experience is often considered vital. When participants in a gamified experience view one another engaging in the same activity and being rewarded for it, they can display and compare progress, giving them a heightened sense of accomplishment, belonging, and validation (Hamari, 2017). It would be interesting for further research to analyze if any summer book clubs are successful at filling this peer-to-peer gap by encouraging a sense of social belonging between the reader and her community or household members.

Storytime and gamified play

Like summer reading programs, official story hours are a long-standing tradition in public libraries, with official librarian training taking place by the early 1900s to better legitimize the practice (Jenkins, 2000). While these early programs aimed to inspire a love for stories and advertise the library, around the 1950s as discussions around early literacy were increasing, they transformed into ways to increase reading and language skills in young patrons (Albright et al., 2009). Today, many focus on accessibility and bringing stories to a wider array of demographics. As Table 3 lists, libraries in this survey typically delivered storytimes in at least English and Spanish, with the majority also offering a range of age-specific reading times for babies, toddlers, and those in elementary. Other types of storytimes also appear to address emotional, physical, and social considerations, for example by creating sensory-aware experiences for children with sensitivities or providing therapy dogs for children to read to in a safe, non-judgmental space. Though in some cases virtual storyhours did not match up with their physical counterparts in terms of quality or quantity, over half of libraries provide live or prerecorded series that encourage energetic, physical engagement.

If there was one common thread between the storytimes offered by libraries in this survey, it's that sitting quietly with your hands under your legs listening to someone prattle on is a thing of the past (Figure 5). In its place are immersive, playful, and hands-on experiences that places the learner, not the teacher, at the center of the learning process. Libraries now frame their storytimes as dialogic activities, meaning that children are highly engaged in the unfolding narrative. In dialogic reading, the main speaker routinely pauses to ask the children questions to pinpoint their understanding, encourage them to discuss amongst themselves and share perspectives, or even work together to imagine a new part of the story. While children are given a more active role, the librarian scaffolds the experience to provide guidance and structure. This parallel dynamic has been shown to support children's sense of competence as it communicates clear expectations and the steps needed to achieve the desired outcome (Naeghel et al., 2016) while increasing autonomy and influence so children feel more inclined to participate, share knowledge, and better process the content (Yousefi & Mirkhezri, 2019). This form of active reading has been shown to play a critical part of the development of children's pre-literacy and early literacy skills (Albright et al., 2009).

The most common tactic observed in the survey to get children involved in this highly autonomous, active, but still scaffolded manner is through guided movement (Figure 6). With offerings such as Step into Storytime, Storytime Yoga, and Rock N Read, storytimes

Figure 5. 1917 photo of children sitting in pews, paying close attention during story hours at the Carnegie Library.
Source: Carnegie Museum of Art Collection of Photographs, hosted by Historic Pittsburg.

Figure 6. A child and caretaker engaging in physical play at a library
Source: Hickey, Golden, and Thomas (2018).

were often described as using rhymes, music, and dancing to encourage young learners to perform, interpret, and communicate meaning in their own playful and self-directed way. Research has shown that movement, along with music, art, and tactile play empower children to process and communicate their understanding even if they don't necessarily have the language yet to phrase their beliefs aloud (Diamant-Cohen, 2007). Children of all ages, including babies and toddlers, can actively engage in early literacy skills through puppets, pantomime, and finger art in a way that encourages their unique creativity and gives them a supported voice. When children listen to and interpret stories in their own way, they are more likely to grow intellectually, creatively, and psychologically (Diamant-Cohen, 2007).

In many of these sessions, children not only experiment with their own identities in self-driven ways, but engage with their peers as well, an important element in fostering internal motivation according to SDT. They read, move, and create in a group-based atmosphere together with other children and adults. Studies have shown that when children become immersed in a social circle that meets their personal needs for belonging and comradery, they feel more intrinsically motivated to continue participating (Rapp, 2017). Additionally, when players feel they have a positive community to perform in, and feel they are in control of their sharing, they become motivated to improve their performance to meet their peers' comparison mechanisms, and even excel compared to their group (Rapp, 2017). In a study conducted by Bowser and Rehm (discussed in Thiel &

Fröhlich, 2017), researchers found that community membership and feeling part of a group were one of the main factors leading to participation and motivation.

Spaces and autonomous exploration

According to pictures, design plans, and written descriptions, the physical spaces analyzed in this survey vary widely – with children's sections taking up corners, to floors, to entire buildings – as do their contents. For example, the Elmhurst Public Library in Illinois has a dedicated space called the Learning Garden where pre-k children and their caregivers are encouraged to use toys and puppets to cooperatively learn, play, build, and clean with one another. Taking things up a notch, the Hillary Rodham Clinton Children's Library & Learning Center is entire 30,000 square foot building that includes a theater, a teaching kitchen, and nature spaces such as a greenhouse and vegetable garden. Perhaps most technology-focused of all is the children's floor within the Carnegie Library of Pittsburgh, which boasts specialized programming rooms, tech labs, and hands-on tools such as robots and coding kits. Which is not to take away from spaces such as the Brentwood Public Library which may not have robots, but does boast magical-inspired architecture and design with leafy trees looming overhead, massively oversized faux books, and glowing archways.

When you compare what children's library rooms looked like historically to how they're now described, it's easy to see components of gamification at work. 'Learning through play' and 'interactive' are common phrases highlighted by libraries when they describe their space and the games, items, or events within them. However, if the lack of literature is any indication, the topic of gamification in children's library spaces isn't a widely explored subject, especially in regards to how the physical layout may foster feelings of autonomy, competence, and social relatedness. Still, this concept seemed critical to analyze. Not only is there a clear intent on the part of libraries to reinvent their youth-specific spaces, but the physical and imaginative layout was one of the major considerations for online writers when ranking the top children's libraries. This suggests that the space is a defining feature for patrons and heavily influences their experience and lasting impression.

In 'Beyond Gamification' (2014) Nolan and McBride define four dimensions that are critically important to creating a successful digital game-based learning experience. These elements, which are autonomy, play, affinity, and space, provide a helpful way to discuss the strategies seen in gamified library spaces. Affinity is, 'the role of interest in particular objects, processes, experiences, and locations that draw or attract us' based on our intrinsic values (p. 598). Game spaces that support affinity offer unique modes of participation, social organization, and content for the player to explore in the ways that interest her. As players, or in this case young library patrons, navigate through the space in ways that interest them and work through authentic tasks, affinity links their active play to learning. To experience this autonomous play and meaning making, individuals must feel as though they can interact with the environment without pre-determined expectations or oppressive adult observation. As cited by Nolan & McBride, leading scholar Elliot Eisner critiques popular planning and design in education, noting that school architecture often restricts how children are allowed to move and interact in the space by removing soft surfaces where children can comfortably sit.

Figure 7. Playful children's book return.

Source: Laramie County Library website. Copyright information: 'The Laramie County Library System has made the content of these pages available to the public and anyone may view, copy or distribute information found here without obligation to the Laramie County Library System.'

Figure 8. 'Stan' the Tyrannosaurus-rex and the 'Spirit of Cerritos' lighthouse in the Cerritos library.

Source: Chris Jepsen for Cerritos Library found at https://www.flickr.com/photos/traderchris/5544299765 Copyright information: Free to copy and distribute with appropriate attribution for non-commercial purposes without modification.

Such design intentionally limits spaces that allow for play and freedom away from adult surveillance and assessment.

Using these definitions set forth by Nolan & McBride, there are several examples of libraries included in this survey who have constructed their children's spaces in ways that prompt independent exploration to encourage authentic learner interest. For example, the Laramie County Early Literacy Center encourages children to explore the space to uncover 'learning spots' to engage in literacy-related activities and tinker with hands-on, imaginative installations (Figure 7). At the Cerritos Library, children pick a seat beneath a towering tree or space shuttle, read a book or learn from a number of exhibits, or just sit and listen to the sounds of a rainforest while admiring the overhead dome cycling through time and weather patterns (Figure 8). Spaces such as these allow children to reorganize, explore, and repurpose components of the environment in ways that feel natural to them. As a result, children with a range of age, emotional, and development needs can successfully engage with the space in ways that are beneficial to their own learning.

Much like balancing learner and teacher responsibility in storytimes, gamification best-practices recommend that exploration should be coupled with guidelines to steer learning in a productive direction and support feelings of competence. Integrating goals into the space is one way of achieving this as they 'give the impression that there is always something to do,' and always something 'not yet accomplished' (Rapp, 2017, p. 455). Importantly, goals should not be pre-defined tasks – rather, they should allow for self-determined routes to completion in ways that empower user choice. The sense that there are tasks to be finished and that they can be completed in whatever way feels enjoyable to the player increases intrinsic motivation and strengthens involvement in the activity. In Laramie's learning spots mentioned previously, each zone is marked by a plaque containing the library's cute mountain lion mascots. These mascots provide an easy-to-read explanation of what the learning activity is, what benefit it has, and what questions to reflect on to gain a deeper understanding. Some signs also recommend books that relate to the activity theme. Plaques such as these set clear goals within the space to add necessary guidance to children's freedom to explore. This helps focus the learner's attention in a specific direction and avoid a scattering of aims (Rapp, 2017).

Conclusion

Though the term gamification wasn't explicitly listed as a guiding force by libraries included in this survey – it is, after all, a relatively new tool and methodology – its components can be clearly seen in many programs and initiatives targeted toward youth literacy. Gamification best practices emphasize encouraging the user journey with authentic rewards, supporting unique ways to make meaning and share that understanding with peers, and balancing individual exploration and learning with guiding goals and aims. By integrating these approaches, library summer reading clubs, storytimes, and physical spaces have the potential to support their young patrons' intrinsic desire to feel in-control, able, and socially connected, in turn motivating them to want to engage, read, and learn because they find these activities enjoyable.

The goal of this paper was to use public messaging to discuss library programs and spaces through the lens of gamification to help define the increasingly popular framework in an approachable manner and highlight its potential positive impact on authentic motivation and learning. By analyzing how libraries are constructing their initiatives to support gamification and intrinsic motivation according to SDT, the authors aim to provide actionable insight for other educators who are interested in applying the methodology to better support young literacy learners. This survey and discussion are a first step toward constructing a foundation of definitions, approaches, and public-facing communication strategies. Future research can build off this base layer to engage in a more informed manner with librarians and educators themselves to gain a sense of how these professionals are discussing gamification principles amongst themselves and their institutions and whether they see a future where gamification plays an even more integral role in how they deliver and market their strategies.

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